

Assignment 2

Studies in various populations have suggested that living in proximity of coal-fired power stations increases the risk of developing chronic obstructive pulmonary disease (COPD), a group of lung conditions that cause breathing difficulties and includes emphysema and chronic bronchitis. Strong evidence indicates that COPD risk is also affected by a series of individual/household-level factors, including age, sex, smoking habits and household use of solid fuel for cooking.

*In preparation for a large-scale national study, you have been asked by the Health Department to provide preliminary data on the **prevalence of COPD among the population 15 years and over, both in towns located in proximity of a power station and towns with no power stations in their territory.** The request also asks for **collecting information about potential confounders that can affect the comparisons in order to plan adequately for future study.***

*As this is a preliminary study with a limited budget and a short time frame, the Department does not require a nationally representative sample, and you are allowed to **select purposely one town with power stations in their territory and one without.** The level of precision requested for the estimates is $\pm 5\%$, with a level of significance $\alpha = 0.05$. From studies in similar populations we expect a COPD prevalence of about 20%.*

Your tasks in this assignment are:

- 1) to write the **aims & objectives, methods** and **ethics** sections of a research protocol for a survey that you deem adequate to answer that request.
- 2) To use the SurveyLab to **conduct the survey according to your protocol**, and provide a short summary of the main results, including:
 - The **estimated prevalence of COPD** in each of the two chosen towns, including their 95% CI;
 - The **estimated prevalence of the potential risk factors** listed above, separately per town (no calculation of CI is necessary);
 - Overall **response rates** at household and individual level;
 - Information on the overall **cost of the survey** (set up, collection, unit cost).

Please, include in the submission **a copy of the data downloaded from the SurveyLab** (the *data.csv* file).

Note: the request is for the **diagnosed prevalence of COPD**, not the self-reported one.